

L4 Runtime Enforcement Profile v0.1

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c Hardening Pack v0.1

c = a + b / L4 / SER / L4 Witness / ARL / Continuity Bundle

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Document class: runtime-enforcement profile / L4 hardening profile / conformance artifact

Assertion class: C-A10 control-layer artifact; C-A7 where witness/log validation is required

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It is not legal advice.

It is not product certification.

It is not implementation-ready.

It is not deployment-ready.

No conformance-supported status is claimed unless separately validated by evidence, tests, and review.

L4 Runtime Enforcement Profile v0.1

Hard and soft runtime enforcement for cost, time, energy, tokens, storage, network, agent mesh growth, retry storms, and self-modification

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0. Executive definition

L4 Runtime Enforcement defines how a c implementation **MUST** convert reality-bound constraints into runtime gates.

It closes a specific criticism:

L4 cannot remain only a philosophy of cost, time, energy, and irreversibility.
It must stop, hold, freeze, quarantine, or degrade runtime behavior when limits are reached.

Compact rule:

No action without budget.
No expansion without witness.
No retry without cost.
No self-modification without gate.
No override without accountable trace.

1. Scope

This profile applies to runtime paths where c or its governed agents may consume resources or create irreversible effects:

- LLM/API calls;
- local CPU/GPU inference;
- tool execution;
- agent spawning;
- retries and background loops;
- memory writes and compaction;
- witness creation;
- network egress;
- file writes;

- physical or financial actions;
- self-modification;
- external handoff or publication.

2. Non-goals

This profile does not:

1. define hardware metering in full;
2. require one vendor or one cloud;
3. replace ARL or L4 Witness;
4. make all failures impossible;
5. authorize autonomous escalation outside scoped privilege;
6. claim industrial maturity without stress evidence.

3. Resource dimensions

Dimension	Minimum meter	Hard stop possible?	Notes
wall_time	monotonic timer	Yes	Required for loops and tasks.
tokens_in/out	provider/local estimate	Yes	Must include retries.
api_spend	price table / account meter	Yes	External oracle must not be infinite.
cpu_time	OS/process meter	Partial	Platform-dependent.
gpu_time	driver/process meter	Partial	At least coarse accounting.
storage_bytes	file/db delta	Yes	Includes logs, memory, witness.
network_egress	allowlist + byte count	Yes	Default deny where possible.
agent_count	live worker counter	Yes	Prevents swarm expansion.
retry_depth	per-task retry count	Yes	Prevents retry storms.
queue_growth	enqueue/dequeue counters	Yes	Prevents background pressure.
witness_cost	record count/size	Yes	Witness itself has L4 cost.
self_modification	patch/write gate	Yes	Requires explicit privilege and witness.

4. Enforcement states

State	Meaning	Runtime behavior
L4-NORMAL	within budget	continue
L4-WARN	approaching budget	reduce scope / prefer local / summarize
L4-SOFT-STOP	soft budget reached	ask, delay, downgrade, or queue
L4-HARD-STOP	hard budget reached	stop current path
L4-HOLD	ambiguous or incomplete budget state	hold until validated
L4-FREEZE	disputed or unsafe path	freeze affected privilege/memory/action
L4-QUARANTINE	suspected runaway/compromise	isolate task/agent/output

State	Meaning	Runtime behavior
L4-RECOVERY	post-stop validation	validate trail, summarize, resume only if allowed

5. Budget classes

Class	Use	Override allowed?
B0	no budget / no action	No
B1	low-risk local diagnostic	bounded local only
B2	ordinary interaction	limited auto-spend
B3	background assimilation	scheduled / throttle required
B4	agent mesh task	task contract + witness
B5	high-cost oracle / external tool	explicit permission or policy gate
B6	irreversible / physical / financial	pre-witness + ARL/L4 check
BX	forbidden	No

6. Mandatory runtime gates

6.1 Pre-execution gate

Before a non-trivial task starts, runtime **MUST** check:

```
task scope
budget class
resource caps
privilege scope
agent count
network policy
memory write permission
witness requirement
self-modification flag
```

If any required field is missing:

```
fail closed -> L4-HOLD or L4-HARD-STOP
```

6.2 Retry gate

Every retry **MUST** consume budget and increase `retry_depth`.

Forbidden pattern:

```
failed task -> invisible retry -> more failed task -> invisible retry
```

Required pattern:

```
failed task -> retry_depth++ -> budget decrement -> stop/hold at threshold
```

6.3 Agent mesh gate

A governed mesh MUST define:

```
max_live_agents
max_enqueue_per_tick
max_retry_depth
max_child_tasks_per_task
max_wall_time_per_task
max_network_egress_per_task
memory_write_allowed=false by default
```

An agent MUST NOT spawn a durable new actor without c-level authorization.

6.4 Witness-cost gate

Witness records are not free.

If witness volume threatens budget, the system MUST reduce frequency, summarize with digest, or close an epoch. It MUST NOT silently stop witnessing while still claiming high-assurance operation.

6.5 Self-modification gate

Self-modification requires:

```
explicit scope
patch diff
rollback plan
witness pre-state
budget
review class
post-state validation
```

No self-modification may silently expand privileges.

7. Gradual resource leakage controls

The profile treats “death by a thousand cuts” as a first-class failure.

Required counters:

- low-cost background tasks per hour/day;
- cumulative token/API spend;
- cumulative witness bytes;
- cumulative storage growth;
- silent queue backlog;
- repeated low-risk calls to the same subsystem;
- tasks reopened after closure;
- orphaned background loops.

If slow leakage exceeds threshold:

```
L4-WARN -> L4-SOFT-STOP -> L4-HOLD -> L4-QUARANTINE
```

8. Required witness event fields

An L4 enforcement event SHOULD include:

```
l4_enforcement_event:
  event_id: string
  created_at: string
  actor_ref: string
  task_ref: string
  resource_dimension: string
  budget_class: string
  budget_before: number | string
  budget_after: number | string
  threshold: number | string
  enforcement_state: L4-NORMAL | L4-WARN | L4-SOFT-STOP | L4-HARD-STOP | L4-HOLD | L4-FREEZE | L4-QUARANTINE | L4-RECOVERY
  action_taken: continue | downgrade | delay | stop | freeze | quarantine | require_review | deny
  override_requested: boolean
  override_granted: boolean
  witness_ref: string | null
  previous_event_hash: string | null
```

9. Conformance tests

Test ID	Title	Required behavior
L4-RT-001	Hard budget exhaustion	Task stops and emits witness.
L4-RT-002	Retry storm	Retry depth stops loop and quarantines if needed.
L4-RT-003	Agent mesh expansion	Excess live/enqueue rejected.
L4-RT-004	Network default deny	Unscoped egress denied.
L4-RT-005	Witness cost overflow	Claim downgraded or epoch closed, no silent witness loss.
L4-RT-006	Self-modification without gate	Patch denied.
L4-RT-007	Death by thousand cuts	Cumulative low-cost leakage triggers hold.
L4-RT-008	Override request	Override requires scope, witness, and accountable actor.

10. Red-line failures

A system fails this profile if it:

1. claims L4 enforcement but has no measured budget state;
2. retries without cost;
3. allows unbounded agent spawning;
4. allows direct memory writes from agents by default;
5. disables witness under pressure while keeping high-assurance claims;
6. treats missing budget state as permission;
7. silently expands privileges during self-modification;
8. lacks a stop state for runaway background work;
9. claims hard enforcement while only logging warnings.

11. Bridge set

Explicit bridge

L4 is the reality boundary of $c = a + b$. This profile translates that boundary into runtime stop, hold, freeze, quarantine, and recovery states.

Quiet bridge I — cybernetics

A regulator that cannot interrupt its own runaway loops is not a regulator. L4 enforcement is the interrupt path that prevents cognition from becoming uncontrolled consumption.

Quiet bridge II — information theory

Every action consumes channel capacity. Budgets force the system to choose what information is worth processing rather than treating infinite context and infinite retries as intelligence.

Earth paragraph

A breaker that only logs “overcurrent detected” while the cable keeps heating is not a breaker. L4 enforcement must actually cut, hold, or isolate the path. Otherwise it is decoration, not safety.

12. Open issues

ID	Issue	Priority
OI-L4RT-001	Extract exact code paths from ester-clean-code.	P0
OI-L4RT-002	Define platform-specific CPU/GPU meters.	P1
OI-L4RT-003	Define token/API pricing registry.	P1
OI-L4RT-004	Create fixture pack for retry storm and slow leakage.	P0
OI-L4RT-005	Bind enforcement events to LHWAI anchors.	P1